Task – version A

Read the description of a pair of identical twins.

Mateo and Felipe are identical twins. Mateo is five minutes older than his brother. Today is their 30th birthday.

'Feliz cumpleaños, big brother.' Felipe can speak fluently in both Spanish and English. Mateo only speaks English but with a strong Scottish accent which sometimes confuses his brother.

When they were very young, family circumstances forced the boys apart. Mateo went to live with his father, a football coach, in Edinburgh, Scotland, whilst Felipe remained with his grand-parents on the family olive farm in Andalucia, Spain.

Both young men have dark brown eyes and black hair. Mateo's hair reaches his shoulders and Felipe's is very short. They have a wide gap between their two front teeth.

When he was a toddler, Felipe used to roll up his tongue and poke it out at his grand-mother's parrot, Loro, which he did not like. Mateo broke his leg when he was eight and learnt from the doctor that his blood group was O+. In Andalucia, when Felipe was told about the accident he decided to send his brother a keyring carved in the shape of a bull. Whilst wrapping the present, he noticed his fingerprints on the sticky tape and wondered for a moment, if they were identical to Mateo's.

When he was seventeen years old, Mateo applied to do sports science at university. He was amazed to learn that Felipe had chosen to do the same course in the same city! They decided to share a flat with two friends. At first, their friends had no problem telling the twins apart as Felipe had very dark skin whilst Mateo's face was pale. Felipe was also one centimetre taller than his brother.

The friends soon decided to join a gym. They were very competitive! It was discovered that Felipe had a greater grip-strength than his twin and could work out for longer on the rowing machine. But Mateo always beat his brother on the squash court.

Today, they both support Manchester United and like to wear beanies when it's cold. Mateo won't admit, even to his twin, that he is afraid of birds. Felipe regularly plays the guitar and their grand-parents describe him as 'the musical one'. Mateo doesn't play any instruments but he does sing opera and Ed Sheeran songs very loudly in the shower!

When you first see identical twins, they can be hard to tell apart. This is because much about the human body is controlled by **genes** and identical twins have identical genes.

As well as (usually) having the same birthday, they also share the same physical features, such as eye colour.

But, as you have read, Mateo and Felipe are not **exactly** the same. How people

live and their **environment**, will alter the effect of their genes.

Instructions

Have 5 different colours ready (they don't have to be identical to those listed below). Choose pens or pencils which make a thin line. These will help you draw around the words you select so you can still read them easily.

Pink – circle all the physical features the twins have. Now draw a table with two columns, one for Mateo and the other for Felipe. Record all their physical features.

Green – In your table, circle the physical features which are different in both boys.

1. Why do Mateo and Felipe have some features which are exactly the same?
2. Can you explain why some of their features are different?

   Brown - circle the coincidences.

   Orange – circle the other skills and attributes they have between them. Draw a second table with two columns and list all the orange-circled words.

   Blue - In both tables, circle any physical features and skills which can be easily measured.
3. Are any of these blue-circled words 'measured' in words or letters (qualitatively)?
4. Using numbers will give you quantitative measurements which can be easily compared. What units would you use to measure the other blue-circled features and skills?
5. Look at your second table. What skills and attributes do you think the twins share and what is your evidence for your conclusion?
6. Which features or skills do you judge to have an equal genetic **and** environmental influence? Explain why you think this.

For further research or home-learning:

a. Can Mateo also roll his tongue? Explain your thoughts.

b. What blood group is Felipe? Outline your evidence.

c. Are their fingerprints identical? Why do you think this?

Task – version B

Read the description of a pair of identical twins.

Mateo and Felipe are identical twins. Mateo is five minutes older than his brother. Today is their 30th birthday.

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When he was seventeen years old, Mateo applied to do sports science at university. He was amazed to learn that Felipe had chosen to do the same course in the same city! They decided to share a flat with two friends. At first, their friends had no problem telling the twins apart as Felipe had very dark skin whilst Mateo's face was pale. Felipe was also one centimetre taller than his brother.

The friends soon decided to join a gym. They were very competitive! It was discovered that Felipe could work out for longer on the rowing machine. But Mateo always beat his brother on the squash court.

Today, they both support Man. United and like to wear beanies when it's cold. Felipe regularly plays the guitar and their grand-parents describe him as 'the musical one'. Mateo doesn't play any instruments but he does sing opera and Ed Sheeran songs very loudly in the shower!

Instructions

Pink – circle all the physical features the twins have. Write a list.

Green – circle the physical features in your list which are different in both boys.

1. What features are the same in the twins?
2. Why do Mateo and Felipe have some features which are exactly the same?
3. Why are some of their features different?

   Brown - circle the coincidences.

   Orange – circle the other skills and attributes they have. Write a list.

   Blue - In both lists, circle anything which can be easily measured.
4. Are any of your blue-circled words 'measured' in words or letters (qualitatively)?
5. What units would you use to measure the other blue-circled words?
6. Look at your second list. What skills do you think the twins share?

7. Why do you think this?

8. Why do you think Mateo is better at squash than his brother?

Teaching notes and suggested answers

The aim of this resource is to promote a better understanding of how physical features and other attributes such as sporting skill are influenced by both genes and the environment. It does this through to prompting discussion, questions and ideas from the students.

The text has been reduced and questions adjusted in version B to open the topic to students who need more support.

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Suggested answer tables

Physical features	Mateo	Felipe	Which can be easily measured?
Colour of eyes	dark brown	dark brown	✓
Colour of hair	black	black	✓
Length of hair	long	short	✓
Teeth	gap between teeth	gap between teeth	✓
Blood group	O+		✓
Tongue roller		yes	✓
Skin colour	pale	dark	✓
Height		+ 1 cm	✓
Grip strength	lower	higher	✓
Stamina	lower	higher	✓

Other skills and attributes	Mateo	Felipe	Which can be easily measured?
Language spoken	English	English and Spanish	
Accent	Scottish		
Sports played	squash	squash	✓
Instruments played	sings	guitar	
	Man united supporter	Man united supporter	
	Wears beanies	Wears beanies	
	Afraid of birds	Doesn’t like the parrot	

Discussion points

1. Why do Mateo and Felipe have some features which are exactly the same?

   These features are controlled by their genes and their genes are identical.

2. Can you explain why some of their features are different?

   These have been affected by their environment: hair can be cut; skin tone is affected by exposure to sunlight and Mateo lives in Scotland whilst Felipe spent his childhood in Spain; grip-strength depends on practice – Felipe has lived on a farm and may have helped out, making his grip stronger. This may also have given him greater endurance. Mateo may be better at squash because this indoor sport is one which can be done regularly in a colder, wetter climate.

3. Are any of these blue-circled words 'measured' in words or letters (qualitatively)?

   Eye colour; hair colour; tongue roll; blood group; fingerprints; skin colour

4. Using numbers will give you quantitative measurements which can be easily compared. What units would you use to measure the other blue-circled features and skills?

   Hair length – cm; gap between teeth – mm; height – m and cm; grip-strength – Kg, N or mmHg (often with a dynamometer); work-out – hours, mins, sec; squash – no unit as such just number of winning games.

5. Look at your second table. What skills and attributes do you think the twins share and what is your evidence for your conclusion?

   Share sporting ability as they can beat each other, although in different sports possibly due to their different exposure to physical activity when young; also father is a football coach, so they will have inherited and share some of his genes. Both like football but why the same team? Is this simply a coincidence?
   May both have a fear of birds – Felipe 'did not like' the parrot; Mateo doesn't admit his fear of birds.
   Both may be equally musical as one plays guitar and other sings opera with confidence.

6. Which features or skills do you judge to have an equal genetic and environmental influence? Explain why you think this.

   Language – humans genetically predisposed to communicate with language.
   Type of language and accent – result of environment.
   Musicality – genes may complement each other to bring about a strong musical potential.
   Instrument and skill – heavily dependent on environment.
   Sporting ability – as above.